

NAIYA DUA

duanaiya02@gmail.com | +91-9691034752 | linkedin.com/in/naiyadua/ | https://github.com/naiyadua

PROFESSIONAL SUMMARY

Final-year B.Tech Computer Science and Engineering student with hands-on experience in Core Java, Object-Oriented Programming, Data Structures and Algorithms, SQL, Python, and backend development. Experienced in implementing Java solutions across arrays, linked lists, stacks, queues, trees, and graphs, with a focus on time and space complexity. Built applications using Python, Flask, SQLite, HTML, CSS, JavaScript, and Machine Learning. Seeking a Software Engineer Intern role to apply problem-solving and software development skills in building scalable and maintainable applications.

TECHNICAL SKILLS

Programming: Java, Python, SQL, JavaScript
Core Computer Science: Object-Oriented Programming, Data Structures and Algorithms, DBMS, Operating Systems, Computer Networks
Backend & Frameworks: Flask
Database: SQLite, SQL
Web Technologies: HTML5, CSS3, Bootstrap
Tools: Git, GitHub, VS Code
Other: Machine Learning, Google Maps API

EXPERIENCE

Data Structures & Algorithms (Java) Intern April 2026 – Present
InternVision

- Engineered Java solutions across 6 core data structures, including Arrays, Linked Lists, Stacks, Queues, Trees, and Graphs.
- Evaluated time and space complexity to optimize algorithm performance.
- Resolved 7 challenging algorithmic problems using Java and DSA techniques.
- Improved code efficiency by 20% through algorithm optimization and performance analysis.

PROJECTS

EduNexa – Educational Institution Management System
Tech Stack: Python, Flask, SQLite, HTML5, CSS3, Bootstrap, Git, GitHub

- Engineered a Flask-based educational institution management application combining 4 CRUD operations, SQLite database integration, and admin authentication.
- Implemented CRUD operations with SQLite to support institution record creation, retrieval, updating, and deletion.
- Configured admin authentication to control access to management functionality.
- Built a responsive web interface using 3 frontend technologies: HTML, CSS, and Bootstrap.
- Used Git and GitHub for source code management and project development.

Automated Fall Detection & Emergency Response Mobile Application
Tech Stack: Python, Flask, Machine Learning, HTML, CSS, JavaScript, Google Maps API, Git

- Built a smartphone-based fall detection system using 2 motion sensors (accelerometer and gyroscope) for real-time monitoring.
- Integrated GPS location tracking and 2 emergency notification channels (SMS and calls) for automated alerts.
- Implemented an “I Am Safe” confirmation feature to reduce false emergency alerts.

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science and Engineering Jain University, Bengaluru
Computer Science and Business Systems (CSE-CSBS) August 2023 – Present

PATENT

Bitcoin Price Volatility – Published Patent

- Contributed to a Bitcoin price forecasting and volatility analysis model using 4 analytical approaches: